



Robust real-time blueberry counting in greenhouses using small-object detection and mamba-driven multi-step trajectory completion

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ABSTRACT

Accurate counting is essential for enabling automated blueberry harvesting and logistics decision-making in greenhouse environments. However, the features of small size, dense distribution, and frequent occlusion of blueberries result in severe missed detections and duplicate counting, making accurate counting highly challenging for yield estimation. To address these challenges, we propose an effective technical framework for real-time accurate counting of ripe blueberries in greenhouse environments, including specific deep-learning algorithms of lightweight small-object detection and mamba-based trajectory completion. First we introduce a hybrid loss function and a multi-scale feature fusion module, both integrated into the YOLOv12n model to enhance small object detection performance. Then in the trajectory prediction phase, we design encoders based on KAN and decoders based on Mamba, integrating multi-target positions and trajectories to achieve efficient and accurate position prediction. Thereafter we put forward a prediction-driven two-stage matching strategy based on ByteTrack, which effectively reduces duplicate counting caused by missed detections. Experimental results demonstrate the validity and robustness of the framework, and also highlight the algorithms' performance advantages. The proposed lightweight small-object detection method increases precision by 8.3% and mAP@50 by 2.9% compared to the baseline on the public PEST24 dataset, and achieves an object detection mAP@50 of 0.877 on the real blueberry greenhouse dataset. The trajectory prediction based on our mamba-based trajectory completion algorithm results in an ADE of 15.4 pixels and an FDE of 24.8 pixels over 6 frames, and a single-frame ADE of 3.68 pixels. For blueberry greenhouse counting task, the overall pipeline achieves an accuracy of 92.5% and a processing speed of 35 FPS on the edge device. It effectively addresses repeated counting caused by occlusion and small target size, thereby achieving reliable fruit detection and accurate counting.

1. Introduction

In facility agriculture, accurate fruit counting forms the foundation for intelligent decision-making, harvest management, and yield prediction. Small berry crops, such as blueberries, are characterized by small fruit size, dense distribution, overlapping ripening periods, and frequent occlusion. These features render traditional manual counting methods inefficient and error-prone, failing to meet the accuracy and scalability requirements of modern automated greenhouse production.

Existing fruit counting methods primarily fall into two categories: traditional image processing techniques and deep learning-based detection algorithms. Representative approaches include threshold segmentation combined with the watershed algorithm [1], image enhancement [2], image segmentation [3], YOLO-based detection [4,5], and methods integrating Transformer and Mask R-CNN [6]. Although

these methods improve the accuracy of frame-by-frame detection, most systems estimate the total count by accumulating frame-wise detections. This approach can easily result in duplicate counting in video-based fruit counting tasks [7]. To address this issue, multi-object tracking (MOT) was introduced to the detection framework to ensure cross-frame consistency by assigning unique identities to each fruit. However, challenges such as trajectory drift and identity switching still persist [8,9].

In real-world scenarios, factors such as camera shake and uneven ground surfaces often result in nonlinear trajectories, leading to common issues such as trajectory drift, identity switches, and duplicate counting [10]. To overcome these limitations, deep learning-based trajectory prediction methods have been proposed, aiming to forecast future object positions based on historical motion patterns. For instance, Trajectory Former [11] and PECNet [12] are capable of modeling complex trajectory dynamics and have demonstrated high-precision

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motion forecasting performance. Nevertheless, relying solely on an individual target's historical trajectory often overlooks the dynamic interactions between agents, and fails to exploit the motion information of neighboring targets. Therefore approaches such as Social LSTM [13] and Social GAN [14] attempt to incorporate the behavior of nearby agents by modeling within-frame interactions. These methods introduce social behavior modeling mechanisms to trajectory prediction, enabling the network to better capture inter-agent motion dependencies. However, such methods focus on local neighborhood relationships but ignore the temporal dynamics of neighboring targets and their influence on the main trajectory. As a result, these approaches struggle to provide effective performance under occlusion conditions. This limitation is particularly critical in small fruit counting, where occlusions are frequent and severe. Currently, mainstream trajectory modeling methods are primarily based on Transformers, LSTMs, and MLPs. However, each has significant limitations in practical applications. MLPs struggle to capture complex nonlinear interactions [15]; LSTMs have limited capability in modeling long-term dependencies [16]; and while Transformers offer strong performance, they require excessive computational resources, making them unsuitable for edge deployment [17]. Therefore, achieving efficient and robust trajectory modeling for small fruit counting under dense occlusion remains a critical and unsolved challenge.

As discussed previously, trajectory modeling using MLPs suffers from limited fitting ability, while self-attention mechanisms incur excessive computational overhead. These drawbacks motivate us to consider more efficient and expressive alternatives. The Kolmogorov-Arnold Network (KAN) [18] can fit high-order nonlinear functions with low computational overhead and minimal dependence on dataset size, making it effective for modeling complex interactions and nonlinear motion patterns between fruits. Mamba [19], a new generation of efficient state-space sequence modeling frameworks, accurately captures long-term temporal dependencies while maintaining linear growth in parameter size and computational complexity. It offers greater representational power than models such as LSTM and incurs significantly lower computational overhead than Transformer architectures. These features are crucial for applications such as greenhouse environments, where edge computing power is limited and target motion patterns

change nonlinearly over extended periods.

To address the challenge of accurately counting densely packed small fruits in greenhouse environments, this paper proposes a comprehensive framework integrating enhanced small-object detection, interactive perception-based trajectory prediction, and multi-target tracking. This framework effectively models complex motion relationships and long-term dependencies among fruits. The proposed pipeline is given as Fig. 1. And our main contributions of this paper are as follows:

- (1) We propose a lightweight small-object detection model based on YOLOv12 [20], with a hybrid loss function (HL) to address the challenges of small-object detection, together with a lightweight module named MCSPF. The proposed method significantly improves blueberry detection accuracy in real-world scenarios and consistently outperforms the baseline on the public small-object detection benchmark PEST24 [21].
- (2) We propose a lightweight GamaKAN encoder that jointly models the historical trajectories and spatial positions of both primary and neighboring auxiliary targets via spatio-temporal interactions. This approach significantly improves trajectory prediction accuracy and robustness under occlusion and complex motion conditions. The framework achieves top performance on our dataset and demonstrates strong generalizability on the MOT17 [22] dataset.
- (3) We put forward a real-time deep-learning based blueberry counting framework that integrates small object detection, spatio-temporal interactive trajectory prediction, and a two-stage trajectory completion strategy. This system achieves high-precision counting in greenhouse environments by effectively reducing duplicate counts caused by occlusion and missed detections.

2. Materials and methods

2.1. Datasets

The blueberry dataset utilized in this study was acquired by an inter-row patrol robot operating in a greenhouse situated in Yunnan Province,

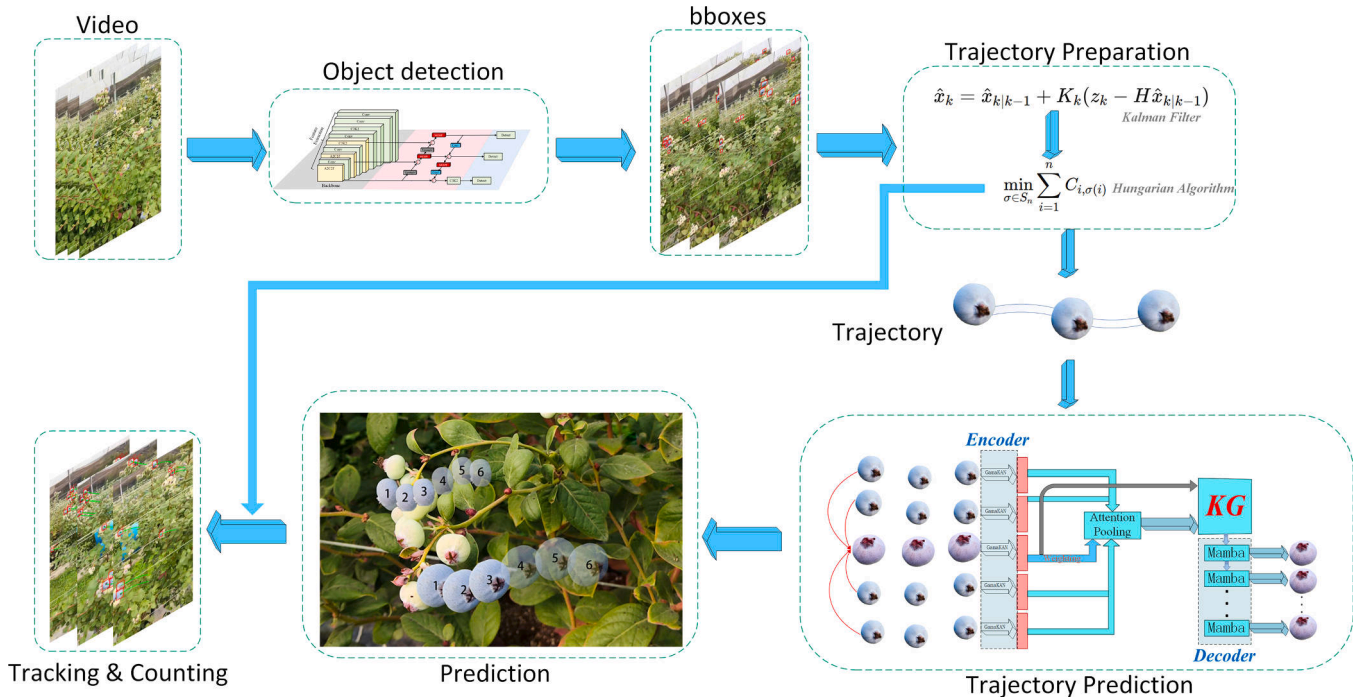


Fig. 1. The proposed tracking and counting framework for blueberry in greenhouses.

China. Thirty-second video segments were reserved for final evaluation, while the remaining footage was sampled at 3 frames per second, yielding a total of 2182 images. Each image was manually annotated with bounding boxes of ripe blueberries using Labellmg. The dataset was split into training, validation, and test sets in a ratio of 8:1:1. On average, 27 ripe blueberries are visible per frame. Additionally, The video dataset is divided into a training set and a validation set. The training set contains 2 min of video with 4132 annotated trajectories of varying lengths. The validation set contains 33 s of video with 1743 annotated trajectories of varying lengths. All videos have a frame resolution of 544×960 and were annotated using Darklabel. A sample from the dataset is shown in Fig. 2.

To evaluate the proposed method on a public dataset, we used the PEST24 dataset, which contains 17,001 images in the training set and 7700 images in the validation set. The dataset contains a large number of small insect instances, presenting significant challenges for detection models. To assess the generalization capability of our trajectory prediction module, we also conducted experiments on the MOT17 multi-object pedestrian tracking dataset. Each frame in MOT17 has a resolution of 1920×1080 pixels. Sliding windows were applied to extract sub-trajectories that meet the input-output sequence length requirements, and each trajectory was converted to NumPy binary format (.npy) for efficient loading. In total, 325,791 valid trajectories were obtained, with sequences from the first half of the videos used for training and the second half reserved for validation.

2.2. Object detection model

Given the dynamic nature of the detection task, the dataset frequently contains challenging samples such as motion-blurred objects, heavy occlusions, and small targets. To address these issues, we adopt the latest YOLOv12 model as the baseline. We propose a hybrid loss

function to replace the IoU loss in the box loss of YOLOv12, and introduce the Small-object Perception Fusion (MCSPP) module to replace the A2C2f module in the head. The enhanced YOLOv12 architecture proposed in this study is illustrated in Fig. 3.

2.2.1. Hybrid loss

Our blueberry dataset contains a large number of small objects, with the smallest target as small as 16 pixels. In small object detection tasks, challenges such as limited target size, blurred boundaries, and amplified localization errors are particularly pronounced. To address these issues, we design a stage-wise hybrid loss strategy that leverages the strengths of different loss functions at different training stages. This method effectively balances global localization accuracy and boundary refinement throughout the optimization process.

Small-Object Focal Wasserstein Loss: During the early training phase, we employ the Small-Object Focal Wasserstein Loss as the primary regression objective. This loss is formulated based on the Wasserstein distance, where the center-point error is normalized by the target's width and height. A focal mechanism is introduced to emphasize regression gradients for small objects and samples with small spatial deviations. In addition, logarithmic smoothing is applied to width and height errors to suppress the influence of extreme scale deviations on gradient magnitude. This stage enables the network to rapidly establish a coarse spatial localization capability and learn scale-aware representations. The formulation of the loss is defined as follows.

$$Loss_w = \text{Focal}(w_{gr} h_{gr} |c_p - c_{gr}|_2) + \log(1 + |w_p - w_{gr}|) + \log(1 + |h_p - h_{gr}|) \quad (1)$$

Hierarchical Boundary Attention Loss (HBA Loss): As training proceeds, the Hierarchical Boundary Attention Loss (HBA Loss) is progressively introduced, with its loss weight incrementally increased. HBA



Fig. 2. Close-up images of blueberries from the blueberry dataset collected in a greenhouse environment. Subfigures (a)-(h) show blueberries at different ripeness stages: green or pink berries are unripe, while blue berries are ripe. In this study, ripe blueberries are the targets for recognition and counting.

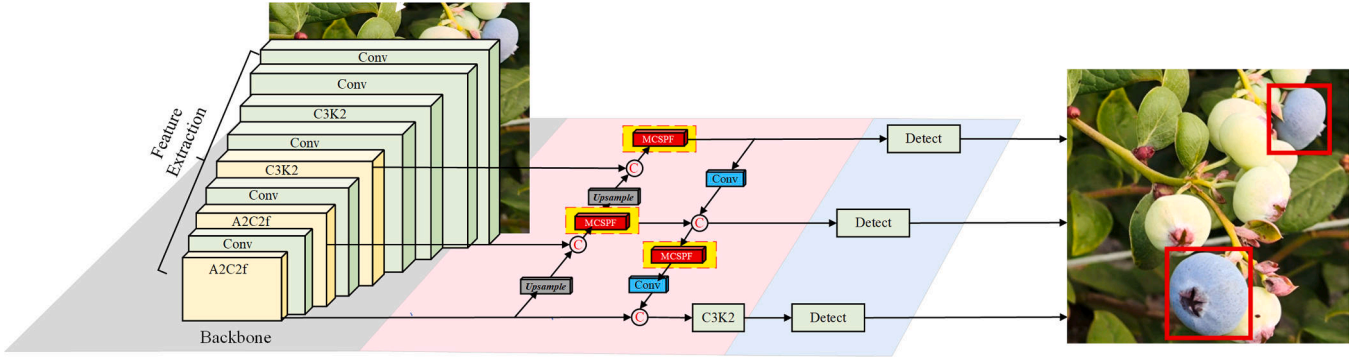


Fig. 3. The improved YOLOv12 architecture, in which MCSPF is the lightweight multi-scale information fusion module developed in this work, represented by a red dashed box on a yellow background.

Loss directly models the normalized errors of the four edges between the predicted and ground-truth bounding boxes. A boundary attention factor is incorporated to adaptively adjust the contribution of each edge during training, enabling the model to focus more on challenging boundary regions. This mechanism enhances the network's capability to regress bounding box edges with higher precision. This stage aims to refine the geometry and completeness of the predicted boxes, thereby improving overall localization accuracy. The computation is defined as follows.

$$\text{HBA Loss} = \sum_{b \in \{l, t, r, b\}} \alpha_b \cdot \frac{1}{w_{gt} + h_{gt}} |b_p - b_{gt}| \quad (2)$$

2.2.2. MCSPF (multi-branch cross-stage pyramid fusion)

To improve the detection of small and densely packed objects, a Multi-branch Cross-stage Pyramid Fusion (MCSPF) module is proposed. Based on the Cross Stage Partial (CSP) architecture, the module divides the input features into two parallel branches: one serves as a direct residual connection to retain low-level semantic information, while the other passes through multiple deep residual blocks followed by a Spatial Pyramid Pooling (SPP) module to achieve multi-scale and multi-level feature fusion.

Specifically, the MCSPF module first applies a 1×1 convolution to divide the input features into two branches. One branch bypasses further processing to retain shallow semantic information, while the other stacks multiple inverted residual blocks for deep feature extraction. An *spp* block is inserted within this path to significantly enhance the network's ability to capture objects at varying scales. The two branches are

concatenated along the channel axis and then fused through another 1×1 convolution to produce the final output.

$$O = \text{Conv}_{1 \times 1}(\text{Concat}[\text{Conv}_{1 \times 1}(X_1), \text{SPP}(\text{ResBlock}(X_2))]) \quad (3)$$

Here, X denotes the input feature map, ResBlock represents the stacked residual block sequence, and SPP refers to the Spatial Pyramid Pooling module. By leveraging multi-branch fusion and multi-scale enhancement, the MCSPF module effectively maintains a trade-off between computational efficiency and feature representation quality. It significantly improves detection accuracy for small objects and in complex scenarios. Moreover, the module demonstrates high inference speed and strong portability, making it well-suited for real-world deployment.

2.3. Mamba-driven trajectory prediction module

The overall structure of the proposed trajectory prediction framework is illustrated in Fig. 4. The module takes as input the multi-frame historical trajectories of blueberry fruits obtained from the object detector and outputs multi-step predictions of fruit positions at future time steps.

First, the OTAM (Other Trajectory Aggregation Module) applies attention pooling to the auxiliary target trajectories, generating contextual features that capture neighborhood interactions. These features, together with the primary trajectory, are then fed into the GamaKAN encoder. The GamaKAN encoder initially employs one-dimensional convolution to capture local temporal dependencies, followed by a channel-wise KAN module to model nonlinear feature

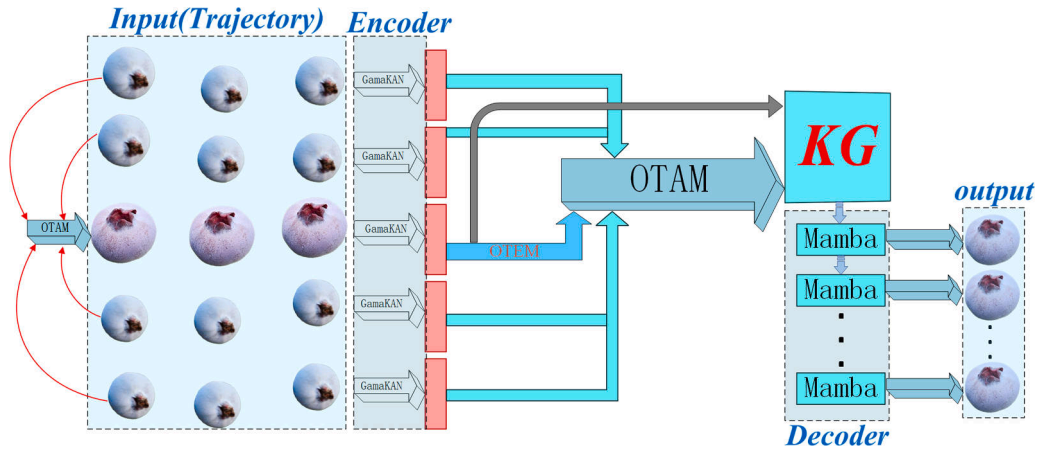


Fig. 4. The structure of the proposed trajectory modeling framework. OTAM aggregates auxiliary trajectories and other agents' positions; GamaKAN encodes the aggregated data; OTEM uses the primary trajectory as reference to perform a weighted fusion of auxiliary trajectory features; the knowledge-graph-fused features are then fed into a mamba-based decoder to predict future trajectories.

transformations. A gating mechanism and residual connection are subsequently incorporated to enhance the stability and expressiveness of the encoded features. The output of this stage is a set of feature vectors encoding both the primary and auxiliary targets.

Subsequently, the OTEM (Other Trackers Encode Model) further captures the dynamic behavior of auxiliary targets and generates temporal semantic features for downstream feature fusion. The KG (Gated Fusion) module performs channel-wise alignment and fusion of the encoded features from both the primary and auxiliary streams, producing a unified high-dimensional trajectory representation via a learnable gating mechanism.

Finally, all fused features are fed into a multi-layer stacked Mamba decoder, which employs an autoregressive strategy to predict the future positions of the primary target over multiple time steps, yielding the final multi-step trajectory predictions.

The overall structure of the trajectory prediction module can be formulated as:

$$\hat{Y} = \text{Mamba}(\text{KG}(\text{GamaKAN}(X_{\text{main}}, \text{OTAM}(X_{\text{aux}})), \text{OTEM}(X_{\text{aux}}))) \quad (4)$$

2.3.1. OTAM (other trajectory aggregation module)

To achieve robust modeling of the primary target in complex scenes, relying solely on its own trajectory is often insufficient to handle occlusions and interactions. Therefore, we introduce an “other-target-aware” mechanism into the trajectory prediction framework by aggregating information from surrounding trajectories to enhance the spatial understanding of the primary target. To this end, we design a lightweight module named Other Trajectory Aggregation Module (OTAM), which models inter-object spatial relationships using attention pooling.

The module takes as input the center-position tensor of all auxiliary targets, where each entry represents the 2D center coordinate of a neighboring object at a given frame. The position of the primary target at each frame is mapped as the query vector, and it interacts with the key and value vectors constructed from the same-frame auxiliary targets via scaled dot-product attention. The output is a spatial context representation that encodes the local environment as perceived by the primary target.

Within the overall framework, the spatial features generated by OTAM are used in two separate paths: (1) they are concatenated with the primary trajectory and fed into the primary trajectory encoder to supplement environmental semantics, and (2) they are aggregated into a global motion representation of auxiliary targets, which serves as input to the auxiliary trajectory encoder. This modular design enables spatial and temporal decoupling and aggregation of primary and auxiliary trajectory information, substantially enhancing the model’s robustness under occlusion and interference.

2.3.2. GamaKAN: lightweight trajectory fusion encoding module

To effectively model the temporal trajectory of the primary target while incorporating its spatial context at each frame, we design a GamaKAN module as the core component of the primary trajectory modeling path. The input to this module is a concatenated sequence composed of the primary target’s historical positions and the context-aware spatial features aggregated from the OTAM module. The output is a dynamic feature representation of the primary target across the entire historical window.

Compared to conventional structures such as MLPs or Transformers, KAN achieves a better trade-off between representational capacity and computational efficiency, making it well-suited for deployment in resource-limited agricultural scenarios. The overall architecture of GamaKAN is illustrated in Fig. 5.

The GamaKAN module first applies a depth-wise separable convolution to encode the fused trajectory sequence. This structure uses channel-wise convolutions to capture intra-channel temporal patterns, followed by pointwise (1×1) convolutions to integrate information across channels. This step generates a feature representation that is both locally sensitive and adaptively responsive to temporal variations, forming a stable foundation for subsequent nonlinear modeling in the temporal dimension.

To enhance the model’s capacity to model complex trajectory evolution patterns, GamaKAN incorporates a nonlinear transformation block based on KAN. According to the Kolmogorov–Arnold representation theorem, any multivariate continuous function can be expressed as a superposition of univariate functions, which theoretically provides

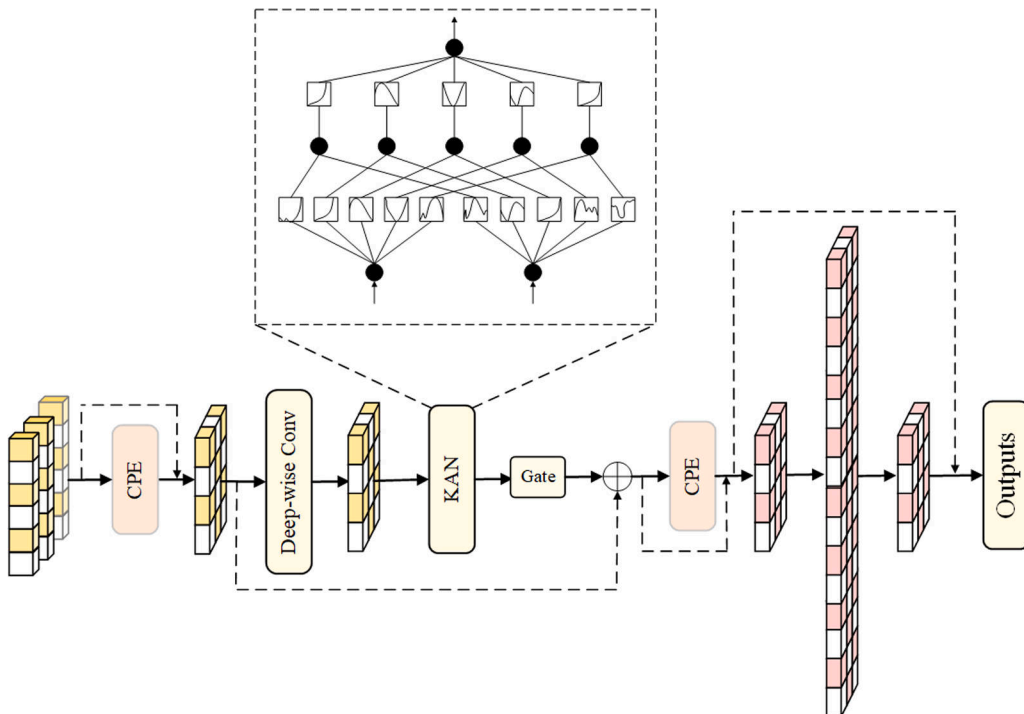


Fig. 5. Architecture of the GamaKAN encoder module.

theoretical justification for channel-wise modeling.

In our design, the KAN module utilizes spline basis functions to implement the nonlinear mappings and applies a linear combination across channels to generate the final output. The core formulation of this computation is as follows:

$$y = W_{\text{base}} \cdot \varphi(x) + W_{\text{spline}} \cdot B(x) \quad (5)$$

Here, $\varphi(x)$ denotes the base activation function, and $B(x)$ represents a set of nonlinear basis functions constructed from splines. W_{base} and W_{spline} are the weight matrices for the base and nonlinear transformation paths, respectively. All operations are performed independently across the channel dimension. This design retains the end-to-end trainability of neural networks while providing strong structural interperability and numerical stability.

To further enhance the stability and robustness of feature representations, GamaKAN introduces a channel-wise gating mechanism after nonlinear modeling. This gating unit learns the relative importance of each trajectory channel for the primary target and adjusts the fusion weight between the nonlinear output and the original input accordingly.

The final output of the GamaKAN module is a high-dimensional temporal representation of the primary trajectory over the historical window. This output is passed to the subsequent primary-auxiliary fusion module, where it is combined with spatial features generated by OTAM. This integration effectively fuses environmental context with dynamic motion cues, forming the foundation for downstream multi-object interaction modeling and trajectory forecasting.

2.3.3. OTEM (other trackers encode model)

In multi-object trajectory prediction, explicitly modeling the motion trends and spatial distribution of auxiliary agents is crucial for enhancing the prediction accuracy of the primary target. To better capture group behavior patterns, we propose a dedicated auxiliary trajectory encoding module that extracts dynamic semantic cues from the surrounding agent clusters and provides rich contextual information for primary target forecasting. The module takes as input a tensor containing multiple auxiliary trajectories, from which n representative auxiliary agents are selected per frame. Their center-point sequences serve as the basis for temporal modeling, balancing computational efficiency and representational completeness.

Structurally, the module adopts an attention pooling mechanism to aggregate the spatial motion features of the selected auxiliary trajectories, where the primary target serves as the query and the auxiliary targets as keys and values. By computing attention weights, the module adaptively fuses auxiliary features, yielding a compact behavioral representation highly relevant to the primary target. The aggregated features are then passed through a lightweight temporal encoder based on the GamaKAN architecture, which integrates linear projection, depth-wise separable convolutions, and KAN-based nonlinear units to extract temporally coherent semantic representations. The design maintains structural symmetry between the primary and auxiliary encoding pathways, facilitating unified modeling and effective feature fusion in subsequent stages.

The output of the auxiliary trajectory encoding module is integrated with the primary target encoding in the gated fusion module to achieve feature-level alignment and unified interaction modeling. This design enables efficient and robust multi-agent behavior modeling, significantly enhancing prediction performance in challenging scenarios such as occlusion. The core computation of this module is summarized as:

$$z_{aux} = \sum_{j=1}^n \alpha_j h_j \quad (6)$$

where α_j denotes the relevance weight between the primary target and the j -th auxiliary target, which is adaptively learned by the gating network, and h_j represents the spatiotemporal feature of the auxiliary target.

2.3.4. Gated fusion module (KG)

To mitigate semantic discrepancies between the encoded features of primary and auxiliary targets—which can cause instability in trajectory prediction when directly concatenated or merged—we introduce a Gated Fusion Module to enable feature alignment and dynamic integration across the two streams.

The inputs to this module are the trajectory encoding features of both the primary and auxiliary targets. Each stream is first projected into a unified representation space via separate linear transformation layers to ensure consistent dimensionality. The projected features are then concatenated along the feature dimension and fed into a gating submodule. This submodule comprises a linear layer followed by a Sigmoid activation, which learns adaptive fusion weights to balance the contributions of each target type.

The fused representation retains the key dynamic patterns of the primary target while incorporating contextual information from the auxiliary group behavior. This mechanism enhances the model's robustness in scenarios involving dense interactions or partial target disappearance, ensuring more stable and accurate trajectory forecasting.

To further enhance the nonlinear expressiveness of the fused representation, we introduce a gated channel-wise nonlinear transformation layer (KG), constructed based on Kolmogorov–Arnold Networks (KAN). This component applies a set of one-dimensional spline-based nonlinear mappings across individual channels, enabling flexible, learnable transformations on the fused feature space. The output of this layer is then passed to the decoder module for final position prediction.

The KG module achieves semantic alignment and integration between the primary and auxiliary encoding pathways. Through its gating mechanism, it provides dynamic control over the contribution of different feature channels, ensuring consistent multi-target modeling and stable trajectory forecasting. The resulting unified semantic representation is forwarded to the decoder to generate multi-frame future trajectory predictions.

2.3.5. Multi-layer mamba decoder

To progressively generate future target positions across multiple time steps, we propose a trajectory prediction decoder based on state space modeling. The core of this module is a Selective State Space Model (SSM), implemented in a multi-layer stacked architecture that performs step-wise prediction along the temporal axis. This design provides high parallelism and strong modeling capacity.

The decoder consists of L stacked SSM decoding units, each responsible for predicting the target's position at one future time step. At layer l , the decoder takes the previous prediction and the encoded trajectory features as input to generate the current output x_l . The prediction process follows an autoregressive scheme, which is defined as follows:

$$x^{(l)} = \text{MambaDecoder}(x^{(l-1)}, f) \quad \text{for } l = 1, \dots, L \quad (7)$$

Each Mamba decoder unit consists of three components: a fixed sine-cosine positional encoding layer, a Selective State-Space Model (SSM), and an output projection module. For each normalized target position input, we apply a fixed sine-cosine positional encoding to transform it into a multi-scale periodic embedding. This enhances the model's spatial awareness without introducing any trainable parameters. The encoding is generalizable across varying spatial scales and sequence lengths.

$$PE_{pi}[d] = \begin{cases} \sin\left(2\pi p_i / 10000^{\frac{2(d/2)}{D}}\right), & d \bmod 2 = 0 \\ \cos\left(2\pi p_i / 10000^{\frac{2(d/2)}{D}}\right), & d \bmod 2 = 1 \end{cases} \quad \text{for } i = 1, \dots, n, d = 0, \dots, D-1 \quad (8)$$

Next, the Selective SSM component is applied to efficiently capture temporal dependencies. This structure supports parallel computation while maintaining strong representational capacity, making it well-

suited for long-sequence modeling tasks. The resulting hidden state y_t at each time step is then projected into a bounding box prediction through an output mapping module.

Specifically, y_t is passed through a fully connected layer to generate the raw prediction, followed by a Sigmoid activation to normalize the output. The predicted trajectory not only serves as supervision for model optimization but also provides real-time target state updates for downstream tracking modules.

2.4. Multi-object tracking framework based on trajectory prediction and two-stage track completion mechanism

In crowded and heavily occluded scenarios, conventional tracking methods that rely on detection and association frequently experience trajectory interruptions and identity switches. To address these problems, we propose a trajectory prediction-centered counting framework consisting of four key components: trajectory state encoding, spatial interaction reasoning, state forecasting, and track maintenance. First, the GamaKAN encoder encodes the historical trajectory of the primary target to generate a dynamic temporal feature representation. Second, the OTAM network models the spatial interactions among auxiliary targets, providing contextual environmental cues for primary trajectory prediction. Third, the primary and auxiliary features are fused using the gated fusion module and passed to the Mamba decoder, which predicts the target's next position. The output not only guides future motion estimation but also serves as a fallback state reference when detection fails during tracking.

To enhance the robustness of trajectory maintenance, we propose a two-stage trajectory completion mechanism built upon the ByteTrack method, which incorporates a trajectory prediction module. In the first stage, the current detection box is matched with all active trajectories using a standard matching approach. During matching, the trajectory state estimate is provided by the linear extrapolation from the trajectory prediction module, and the cost matrix is calculated using the pixel distance between center points. For unmatched trajectories, if their history length meets the minimum requirement of the prediction module, the extrapolated box is activated and participates in the second-stage matching.

In the second stage, the predicted extrapolated boxes are matched with the remaining unmatched detection boxes. If a match is found, the track is updated with the predicted position and continues to be maintained. If a track fails to match any detection for a specified number of consecutive frames, it is considered terminated and removed.

Compared with traditional detection-based multi-object tracking methods, the proposed two-stage completion mechanism incorporates trajectory prediction into the association process. This integration enables the algorithm to utilize both detection and prediction information, which significantly reduces ID switching and trajectory fragmentation caused by short-term occlusion or detection failure, and effectively mitigates repeated counting resulting from frame loss.

As shown in Fig. 6, the area within the yellow box experienced missed detections due to small object size and motion blur. However, when the object was re-detected, the ID remained consistent, demonstrating the robustness of the dual-stage trajectory recovery method proposed in this study.

2.5. Loss supervision

To enhance the detection performance for small objects, we reuse the Wasserstein distance-based loss function described in Section 2.2.1, and employ it to supervise trajectory prediction with the aim of improving small-object trajectory prediction accuracy. In addition, ADE loss, FDE loss, and Velocity Loss (VEL Loss) are combined in a weighted manner to jointly supervise bounding box regression. Experimental results demonstrate that this combination of losses achieves superior performance on small-object scenarios in our greenhouse dataset.

3. Results

3.1. Experimental platform and evaluation metrics

The experiments were conducted on a system running Ubuntu 22.04, equipped with an NVIDIA GeForce RTX 4090D GPU. The implementation was based on PyTorch 2.4.0 with CUDA 11.8 and TensorRT 8.6.1 for acceleration.

For evaluating the object detection module under complex

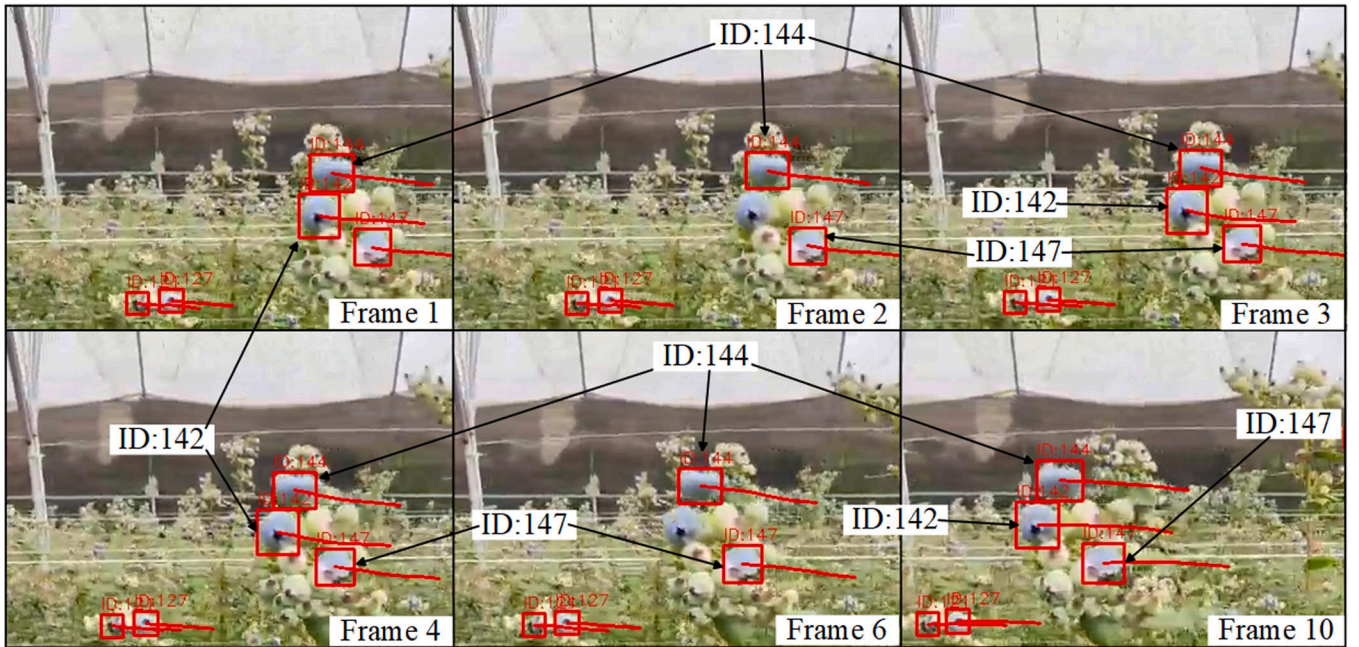


Fig. 6. This figure illustrates the tracklet recovery process in cases of missed detections caused by various factors. Specifically, the blueberry with ID 142 was missed in frames 2 and 6; however, these missed detections did not result in duplicate counting. The stable tracking of IDs 144 and 147, among others, further demonstrates the robustness of our approach.

environmental conditions, we adopt Precision, Recall, and mAP@50 as performance metrics. Precision measures detection accuracy, while Recall reflects the rate of missed detections. The F1-score is used to measure the harmonic mean of Precision and Recall. mAP@50 is a widely used metric that quantifies the average precision at an IoU threshold of 0.5. Additionally, frames per second (FPS) is reported to evaluate the system's inference speed and deployment feasibility.

For the trajectory prediction task, we employ Average Displacement Error (ADE), Final Displacement Error (FDE), and FPS as evaluation metrics. ADE represents the average error between predicted and ground-truth center points over all time steps, while FDE captures the error at the final frame. Lower values for ADE and FDE indicate higher prediction accuracy. The tracklet recovery metrics are further detailed in Section 3.4 and the counting accuracy is calculated as follows:

$$\text{Counting Accuracy} = 1 - \frac{|\text{Ground Truth} - \text{Prediction}|}{\text{Ground Truth}} \quad (9)$$

3.2. Performance evaluation of object detection models

3.2.1. Ablation study of object detection models

To evaluate the effectiveness and generalizability of the proposed Hybrid Loss and MCSPF module for small-object detection, we conducted experiments on the public small-object dataset PEST24 as well as our self-collected blueberry dataset. The ablation results on the PEST24 dataset are summarized in Table 1. In this setting, Loss1 refers to the Focal Wasserstein Loss, and Loss2 denotes the Hierarchical Boundary Attention Loss (HBALoss). The Split column indicates whether the weight of HBALoss is scheduled to increase progressively during training.

3.2.2. Comparison analysis of object detection models

To balance both the real-time performance and detection accuracy required for tracking, we adopt the YOLO series as the detection backbone. To further validate the effectiveness of the proposed detection components, we compare our method against several mainstream YOLO variants, which serve as baselines. The results are presented in Table 2.

As shown in the table, the selected baseline model achieves the best performance on our dataset, particularly with notable improvements in Precision and mAP@50. These results demonstrate the strong capability of our proposed Hybrid Loss and MCSPF module in enhancing small-object detection performance.

Fig. 7 presents the qualitative detection results of different models under challenging conditions, including occluded targets, small objects, and partially visible blueberries. The results demonstrate that our proposed method achieves robust recognition performance across a variety of complex scenarios, effectively addressing the challenges posed by real-world greenhouse environments.

3.3. Performance evaluation of multi-object tracking algorithms

Ablation and comparative experiments are conducted to evaluate the effectiveness of the proposed trajectory prediction module and multi-object tracking framework. The experimental setup is carefully designed to isolate and verify the contribution of each module, demonstrating the superiority of our approach through comprehensive

Table 2
Contrast experiment.

algorithm	Precision	Recall	F1	map@50	GFLOPs
Yolov5	0.782	0.763	0.772	0.812	15.4
Yolov8	0.747	0.751	0.749	0.795	9.1
Yolov10	0.778	0.774	0.776	0.814	8.4
Yolo11	0.790	0.780	0.785	0.828	7.0
Yolov12	0.804	0.795	0.799	0.841	6.0
OURS	0.822	0.811	0.816	0.877	5.8

evaluation. As shown in the Fig. 8, the schematic diagram of multi-object tracking shows that our method in this study performs stably in the real blueberry greenhouse scenario, and the repeated counting caused by ID switching and other factors is significantly reduced.

3.3.1. Ablation study on multi-object tracking algorithm

To systematically assess the effectiveness of each component in our framework, we conducted seven groups of ablation experiments as follows. Group A investigates the impact of the KAN structure and gating mechanism within the GamaKAN module by replacing KAN with MLP, multi-head attention, and linear attention, as well as comparing different gating strategies. Group B replaces the GamaKAN encoder with a Mamba encoder to evaluate the effect of the encoder architecture. Group C compares various pooling methods. Group D examines the influence of removing information from other trajectories and auxiliary target positions, respectively. Group E evaluates the KG module by replacing KAN with MLP and attention mechanisms, and by ablating the gating mechanism. Group F investigates different decoder structures by comparing Mamba, GRU, and LSTM. Finally, Group G represents the complete method proposed in this work, integrating all optimized modules and achieving the best overall performance. Table 3 reports the results in terms of Average Displacement Error (ADE), Final Displacement Error (FDE), and Frames Per Second (FPS).

Results indicate that while baseline methods such as MLP and average pooling achieve high inference speeds, they exhibit large prediction errors, rendering them unsuitable for real-world deployment. Although multi-head attention improves prediction accuracy, it substantially increases computational cost, thereby limiting practical applicability. Removing either the cooperative target modeling or tracking information improves speed but leads to a notable drop in accuracy and generalization ability.

Only the full proposed architecture (GamaKAN + APR + OT + KG) achieves the best overall balance, outperforming all other configurations in both ADE and FDE while maintaining a high FPS. These results confirm the necessity and effectiveness of our collaborative module design for robust and efficient multi-object trajectory prediction.

3.3.2. Comparative analysis of multi-object tracking algorithms

We select several representative state-of-the-art multi-object tracking and trajectory prediction methods, including Social-LSTM, GTR, and MOTR, as baselines for comparison. Evaluation metrics include First Frame Distance (FFD), Final Displacement Error at 6 frames (FDE@6), and Frames Per Second (FPS). The results are summarized in Table 4.

As shown in the table, our proposed method outperforms all baseline

Table 1
Ablation experiment.

Loss1	Loss2	Split	MCSPF	Precision	Recall	map@50	GFLOPs
×	×	×	×	0.584	0.539	0.534	6.0
√	×	×	√	0.660	0.532	0.553	5.8
√	×	×	×	0.593	0.541	0.536	6.0
×	√	×	×	0.617	0.538	0.542	6.0
√	√	×	√	0.621	0.540	0.558	5.8
√	√	√	√	0.667	0.543	0.563	5.8

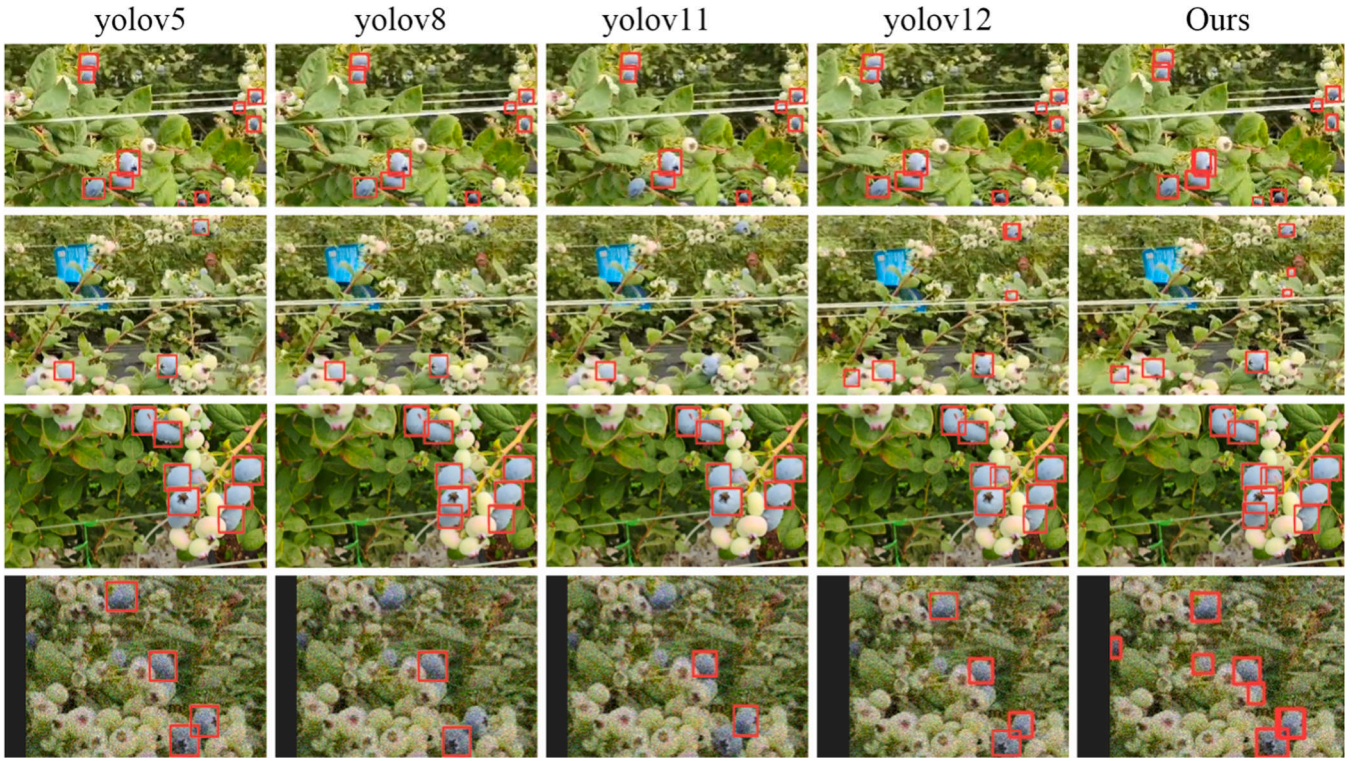


Fig. 7. Detection results of different object detection models.

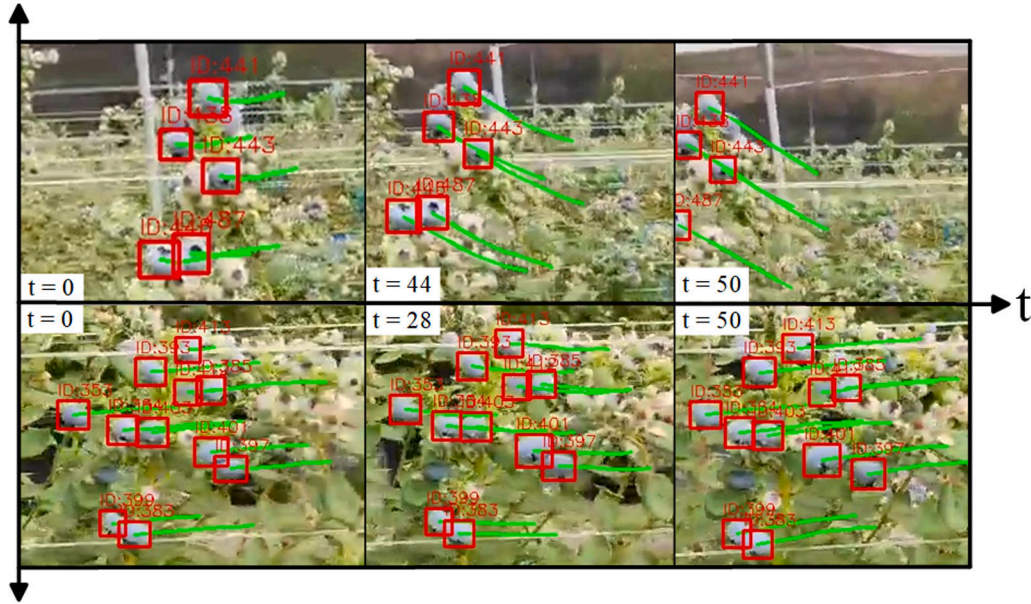


Fig. 8. Tracking results illustration. During long-term tracking, the blueberry targets frequently changed their movement directions and experienced partial occlusions, with extremely dense target distributions within single frames. Nevertheless, the IDs of all targets remained consistent throughout, demonstrating the overall system's tracking robustness and stability under complex scenarios.

models across all prediction metrics. Notably, it achieves a 10.1-pixel reduction in FDE@6 over the second-best method, demonstrating superior robustness and accuracy in position forecasting. Moreover, this performance is achieved while maintaining a high inference speed, highlighting the method's practical applicability and efficiency.

As shown in Fig. 9, our method also demonstrates superior performance compared to TrackFormer and TADAM, consistently outperforming them across multiple evaluation metrics.

3.4. Multi-frame prediction and missed detection completion

To further assess the model's capability in handling short-term frame loss and occlusion, a simulated occlusion test set is constructed by deliberately omitting detection results for selected targets over several consecutive frames. This setup evaluates whether the method can effectively achieve prediction-based recovery. Experimental results show that when detection boxes are omitted for three consecutive frames, the proposed method achieves a trajectory recovery success rate

Table 3
Ablation experiment of trackers prediction.

Group	Variant	ADE(px)	FDE(px)	FPS
A(Gama)	MLP	21.2	38.2	115
	Multi-head Attention	16.1	25.4	52
	GSA	19.8	36.3	83
	-Gate	16.7	29.5	93
	Linear Attention	18.2	33.1	95
B(Encoder)	Mamba Encoder	25.7	46.5	74
C(Pooling)	Grid Pooling(without mask)	27.0	52.4	112
	MLP Pooling	20.7	37.6	103
	Mean Pooling	26.5	50.4	113
	Attention Pooling	20.2	34.7	96
	Tempor Attention Pooling	25.1	46.2	93
D(Others)	Multi-head Attention Pooling	19.1	30.6	56
	-Other Targets	24.3	43.8	110
	-Other Trackers	27.6	50.0	117
E(Gate Fusion)	Multi-head Attention	24.6	39.4	76
	MLP	28.5	54.2	96
	Temporal Attention	25.9	48.3	87
	-KAN	29.0	56.3	93
	-Gate	19.2	32.1	91
F(decoder)	GRU	19.0	29.9	90
	LSTM	17.4	29.5	86
G(OURS)	GamaKAN+APR+OT+KG	15.4	24.8	91

Table 4
Comparative experiment of trackers prediction.

	First frame dist (pixel)	FDE@6 (pixel)	FPS
Kalman filter	13.7		250
GTR	5.10	41.3	45
MOTR	4.71	34.9	26
Social LSTM	8.33	52.4	75
OURS	3.68	24.8	91

of 74.8 %.

Specifically, detection results are manually removed to measure the system's ability to recover missing trajectories. The results are presented in Table 5. Given the density of target scenes, a practical constraint is imposed: if a target is occluded for more than three consecutive frames, it is considered to have exited the scene and is excluded from the recovery evaluation.

In addition, we designed a counting experiment to verify the superiority of the proposed method in the blueberry counting scenario. ByteTrack was selected as the baseline model, and comparative

experiments were conducted with OCSORT and SORT. We report Accuracy, undercounting, overcounting, re-association recall, re-association precision. The undercounting metric reflects missed detections, assignment of the same ID to multiple targets, and similar issues. The overcounting metric represents cases such as missed detections and failed re-identification leading to duplicate counting, among others. The accuracy metric indicates the overall counting precision.

The recall metric for tracklet association measures the proportion of true associations correctly identified by the algorithm, reflecting its sensitivity. The precision metric quantifies the proportion of correctly associated cases among all associations identified by the algorithm, indicating its accuracy in tracklet association. The calculation methods can be expressed as follows:

$$Re - Association - recall = \frac{TP_{RA}}{TP_{RA} + FN_{RA}} \quad (10)$$

$$Re - Association - precision = \frac{TP_{RA}}{TP_{RA} + FP_{RA}} \quad (11)$$

where TP_{RA} denotes the number of successful re-associations after true track fragmentation, FN_{RA} denotes the number of failed re-associations, and FP_{RA} denotes the number of re-associations with incorrect identities. The results are presented in Table 6.

A total of 1743 trajectories were included in the validation set. As observed in the table, SORT lacks a tracklet complementation mechanism, resulting in a high number of missed detections and duplicate counts. Although OCSORT achieves high recall for tracklet complementation and maintains a low number of duplicate counts, its overly aggressive strategy—disregarding spatial distance—often causes some IDs to disappear on the left side of the frame and then be reassigned to new objects entering from the right, leading to a substantial number of uncontrollable undercounting and extremely low precision. While ByteTrack achieves high precision in tracklet complementation, its naive linear extrapolation based on the Kalman filter is insufficient for handling nonlinear trajectories, indicating significant room for

Table 5
Recovery performance under varying occlusion durations.

Occlusion Duration (N)	DE (pixel)	Recovery Rate(%)
1	3.68	90.2
2	9.88	85.4
3	15.21	74.8

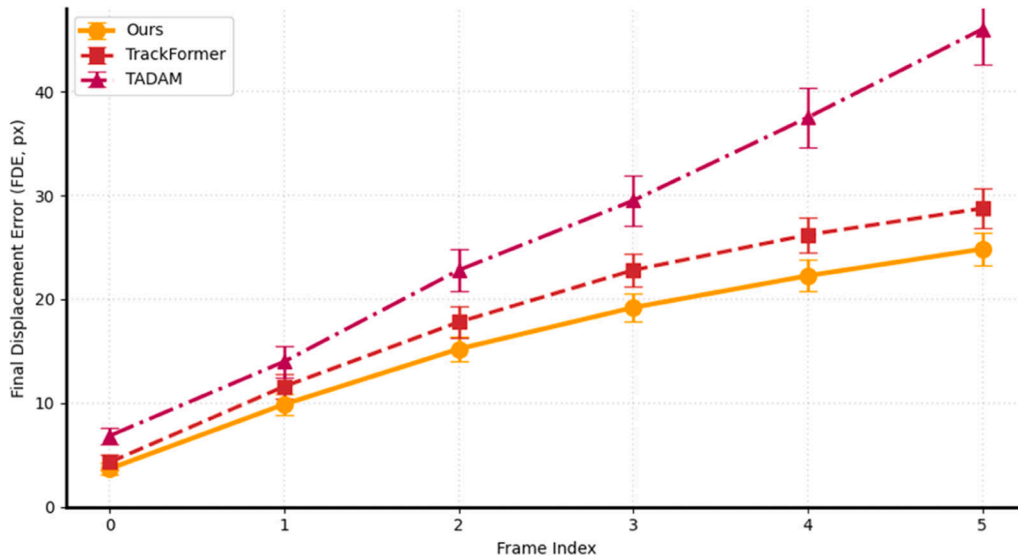


Fig. 9. Final displacement error increases with the number of predicted frames.

Table 6
Comparison of counting results.

Algorithms	Counting	Accuracy	Undercounting	Overcounting	RA-rec	RA-pre
SORT	2031	83.5 %	9.1 %	25.6 %		
OC-SORT	1293	74.2 %	35.7 %	11.6 %	75.1 %	52.0 %
ByteTrack	1902	90.9 %	7.9 %	17.0 %	68.7 %	81.7 %
OURS	1874	92.5 %	3.7 %	11.2 %	73.0 %	87.2 %

improvement.

We further evaluated the overall speed of our method on the Nvidia Jetson AGX Orin 32 GB. The object detection model was accelerated using TensorRT, and the trajectory prediction model was quantized to 8 bits. The system achieves up to 35 FPS, meeting real-time operation requirements. During execution, the process utilized approximately 1.4 GB of system memory and 1518 MB of GPU memory.

3.5. Multi-frame prediction experiments on the MOT17 public dataset

In addition to the above experiments, we conduct evaluations on the MOT17 public dataset to further validate the model's generalization capability and predictive accuracy of our proposed model. This experiment also provides reference metrics for future studies.

We split the dataset by using the first half of the sequences for training and the second half for validation. A 6-frame historical trajectory is used as input, and the model performs step-wise prediction of target positions over the next 6 frames. The final results show a Mean ADE of 26.9 pixels and a Mean FDE of 45.8 pixels, as summarized in Table 7.

Since the evaluation is not conducted on the official MOT17 test set, we do not directly compare our results with those of other methods. Nevertheless, these results serve as a useful benchmark for future studies that adopt the same custom split strategy.

4. Discussion

Results demonstrate that the proposed unified detection–interaction–prediction framework effectively addresses the challenges of high-density, small-object scenarios in greenhouse environments. By leveraging semantically-aware pooling, the method recovers from severe occlusions by aggregating contextual cues from neighboring trajectories. The GamaKAN encoder captures complex, nonlinear motion patterns without compromising inference speed. The multi-layer Mamba decoder then yields smooth, long-range trajectory forecasts, enabling stable object counting even under rapid motion variability. Quantitatively, the proposed method achieves a detection mAP@50 of 0.877 on the in-house blueberry dataset, and improves Precision by 8.3 % and mAP@50 by 2.9 % on the PEST24 small-object benchmark. Furthermore, in real-world scenarios, our method achieves a counting accuracy of 92.5 %, with an edge device deployment efficiency of 35 fps. These results validate both the accuracy and real-time efficiency of our method.

Despite the encouraging results, this study still presents certain limitations and outlines several promising directions for future research. Most publicly available agricultural image datasets focus on specific scenarios or a single object scale, and there remains a lack of high-quality datasets that simultaneously cover both large and small blueberry instances. To better evaluate model robustness under extreme occlusion and small-object conditions, we adopted the PEST24 small-object detection benchmark as a public validation dataset to assess

generalization on challenging samples. While the proposed framework demonstrates strong adaptability across various fruit types and agricultural environments, it has not yet been validated in certain deployment scenarios, such as UAV-based fruit counting.

In addition, the current system primarily relies on RGB imagery for blueberry ripeness estimation, which makes it vulnerable to interference from strong illumination, specular reflection, and occlusion. This limitation particularly affects the ability to distinguish between unripe and discolored ripe fruits, thereby reducing the reliability of maturity assessment. To enhance robustness under such complex lighting conditions, we plan to incorporate multimodal sensing, such as near-infrared (NIR) imaging, to enable more stable fruit state recognition. Moreover, in high-density scenes with frequent occlusions and complex motion trajectories, long-term identity preservation accuracy tends to decline. To address this issue, we propose to fuse static visual cues of fruits (e.g., shape and texture) with dynamic trajectory information, in order to improve trajectory continuity and identity consistency, especially under extended occlusion or intermittent detection loss.

5. Conclusions

This paper presents an effective technical framework for accurate blueberries counting in greenhouse scenarios, featuring in deep learning algorithms of lightweight object detection for small size blueberries and mamba-based trajectory completion for target position prediction in continuous multi-frame scenes. detection, trajectory prediction, and tracking-based counting. For object detection, a hybrid loss function and the MCSPP module are introduced to address the challenges posed by the small size and low detectability of greenhouse blueberries. For trajectory prediction, a Mamba-driven position prediction model is employed to accurately model complex nonlinear movement patterns. And a two-stage tracking and track-completion strategy is implemented to effectively reduce repeated counting for accurate counting task. Extensive experiments on real greenhouse datasets and public benchmarks demonstrate that the proposed method achieves real-time performance and surpasses existing methods in accuracy. These results confirm the framework's applicability for practical deployment in agricultural environments and its strong cross-scenario generalization. Future work will explore cross-modal fusion and multi-task perception to further improve robustness in diverse practical applications.

Ethical statement

The blueberry detection and tracking dataset used in this study was collected by the research team. The dataset contains only images of blueberry plants and fruits. It does not involve human or animal subjects and contains no personal or private information. Therefore, ethical approval was not required for this study.

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CRedit authorship contribution statement

Naiqi Zhang: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Jianhua**

Table 7
Multi-frame trajectory prediction performance on the MOT17 dataset.

Dataset	Future	Resolution	FDE@1	Mean ADE	Mean FDE
MOT17	6	1920×1080	9.0 pixels	26.9 pixels	45.8 pixels

Cao: Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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